



Basic Principles of Medicine 2

Module: Endocrine and Reproductive System | DLA

Autonomic Innervation of the Female Reproductive and Pelvic Organs

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Objectives

SOM.MKII.BPM2.1.ER.1.ANAT.1049	Describe the origin, and type of nerve fibers, found in the superior hypogastric plexus and the inferior hypogastric (pelvic) plexus
SOM.MKII.BPM2.1.ER.1.ANAT.1120	Describe the autonomic innervation of the female pelvic organs and the relationship to the pelvic pain line
SOM.MKII.BPM2.1.ER.1.ANAT.1039	Explain the differences in sensory innervation of the pelvic part of the vagina versus the perineal part of the vagina.
SOM.MKII.BPM2.1.ER.1.ANAT.1128	Compare and contrast the visceral pain pathways for structures above the peritoneal floor of the pelvis versus structures below the peritoneal floor (pelvic pain line) in females



Autonomic functions of the pelvic organs

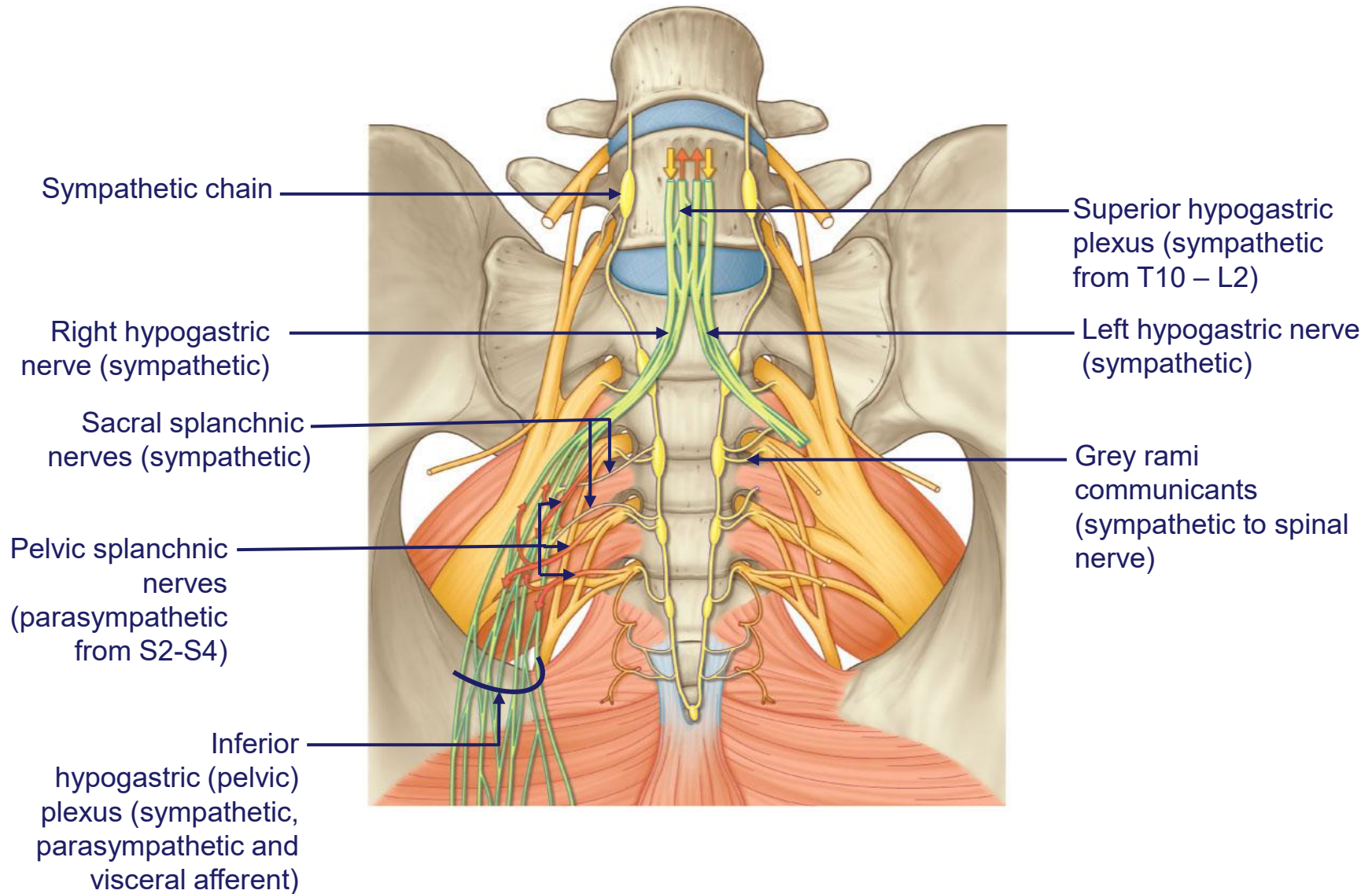
Sympathetic functions:

- Contraction of smooth muscle in blood vessels (vasoconstriction)
- Sphincter constriction - internal urethral and internal anal maintaining continence
- Stimulates uterine contraction
- Promotes vaginal and uterine vasoconstriction
- Relaxation of detrusor muscle of the bladder to promotes bladder filling

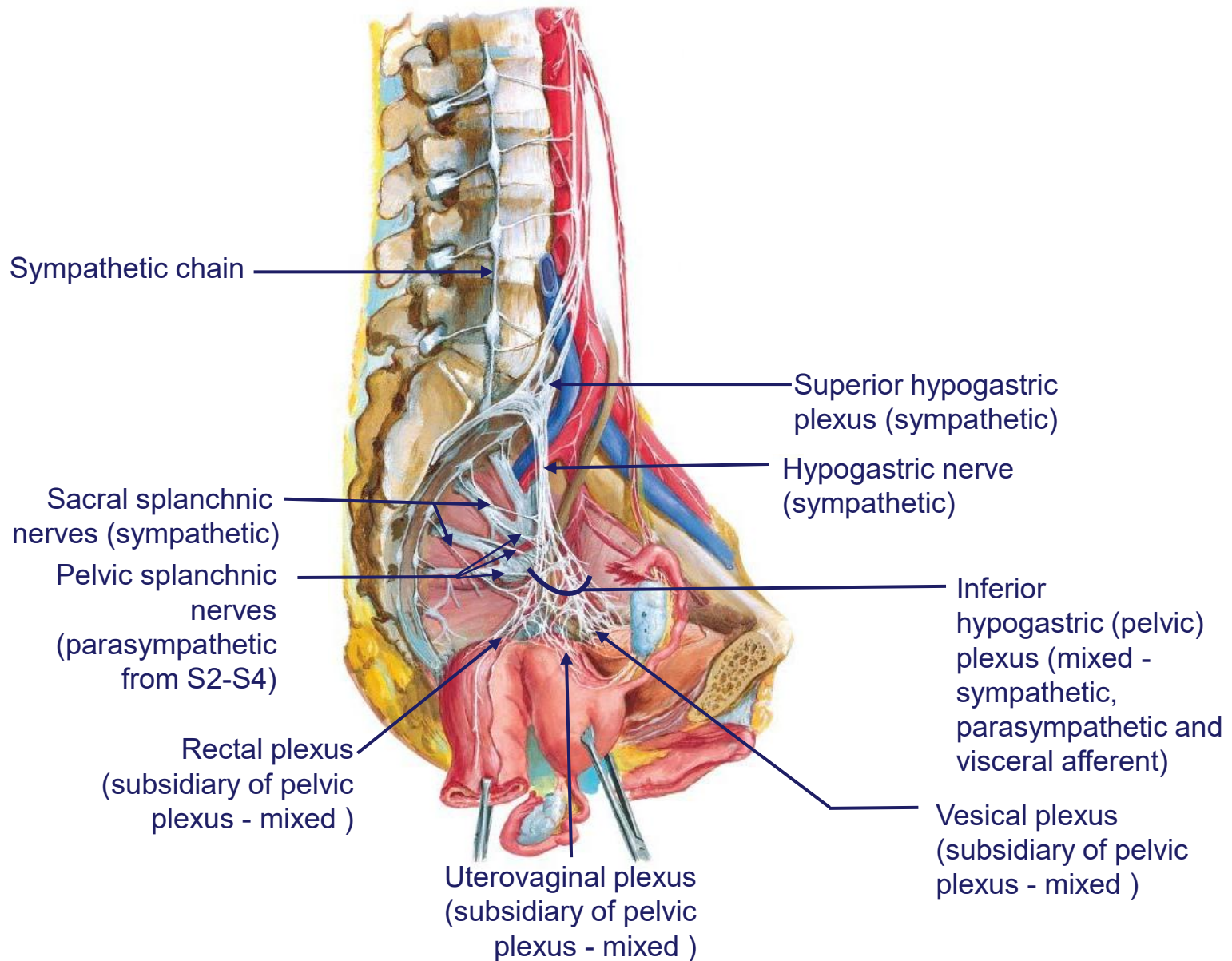
Parasympathetic functions:

- Elicits vasodilation of erectile tissues in the clitoris, leading to erection (via the cavernous nerves).
- Stimulates glandular secretions
- Stimulates contraction of the detrusor muscle, promoting micturition
- Sphincter relaxation - internal urethral and internal anal – promoting micturition and defecation
- Stimulates peristalsis in the rectum, aiding in defecation
- Facilitates vaginal lubrication

Autonomic innervation of pelvic organs



Autonomic innervation of female pelvic organs



Autonomic innervation of pelvic organs

- The autonomic nerves of the pelvis form a complex network that is primarily responsible for the regulation of the pelvic organs, including the bladder, rectum, and sexual organs
- These nerves include both sympathetic and parasympathetic fibers, which often have opposing actions

1. Superior Hypogastric Plexus (Predominantly Sympathetic): This plexus is located anterior to the lower part of the abdominal aorta and bifurcation. It gives rise to the hypogastric nerves

2. Hypogastric Nerves (Sympathetic): These nerves descend from the superior hypogastric plexus along the sides of the rectum to join the pelvic plexuses

Autonomic innervation of pelvic organs

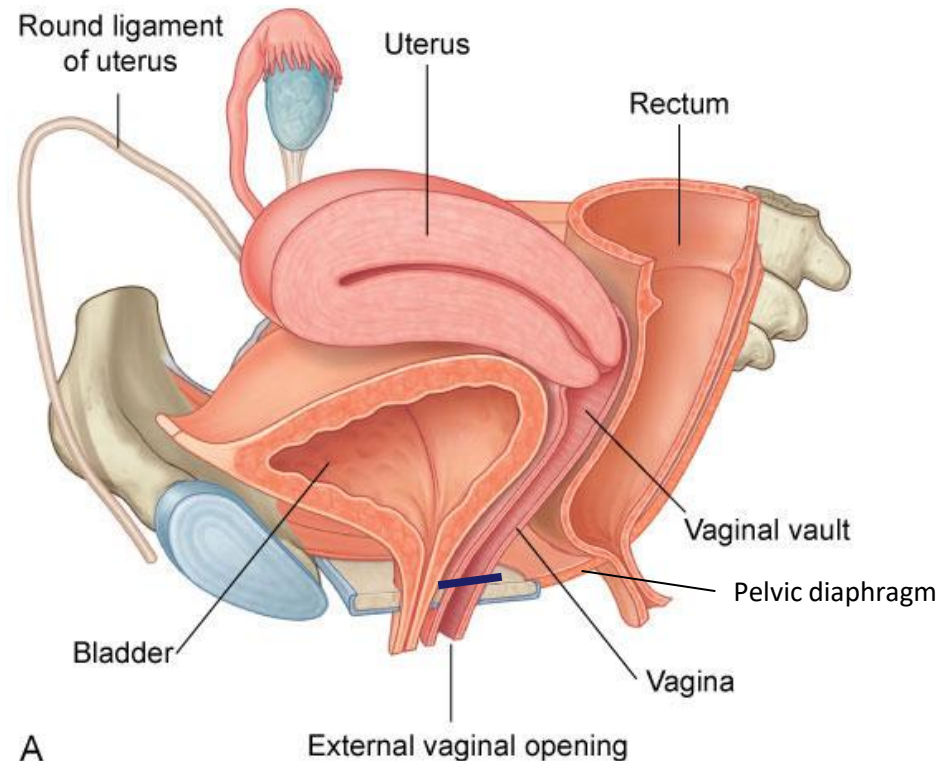
3. Pelvic Splanchnic Nerves (Parasympathetic): These nerves arise from the S2-S4 spinal segments and carry parasympathetic fibers to the pelvic organs

4. Sacral Splanchnic Nerves (T10-L2) (Sympathetic): These nerves arise from the sacral part of the sympathetic trunk, and they carry sympathetic fibers to the pelvic organs

5. Inferior Hypogastric Plexus (Mixed Sympathetic and Parasympathetic): Also known as the pelvic plexus, it lies on the side wall of the pelvis, in front of the piriformis muscle and at the sides of the rectum. **It receives the hypogastric nerves, the sacral splanchnic nerves (sympathetic) and pelvic splanchnic nerves (parasympathetic).** Subsidiary plexuses from this plexus innervate the bladder, rectum, sex organs, and other pelvic structures

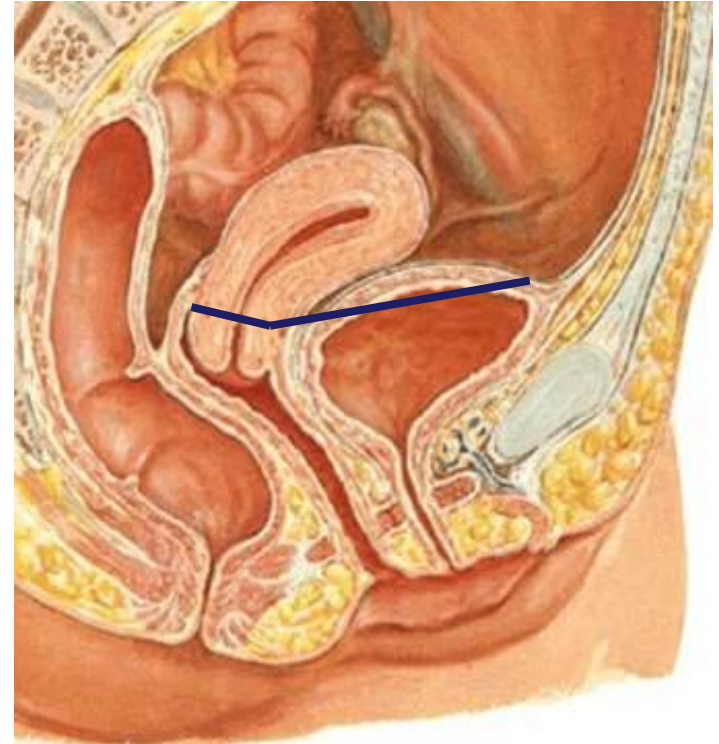
Autonomic and somatic innervation of the vagina

- The vagina has both autonomic and somatic innervation:
 - The upper two-thirds (pelvic portion) receives autonomic innervation through the uterovaginal nerve plexus (containing sympathetic, parasympathetic and visceral afferent fibers),
 - The lower third (perineal portion) receives somatic innervation via the pudendal nerve, specifically its branch, the deep perineal nerve



Pelvic pain line (PPL)

- Imaginary line corresponds to the lower limit of the peritoneum
- **Above PPL (peritoneal part of bladder, uterine tubes, uterine body and fundus)** the visceral afferents for pain travel via the **sympathetics**
 - They travel in the inferior hypogastric plexus
 - Pain is referred to the **T10-L2** areas
- **Below PPL (cervix, majority of urinary bladder, upper 2/3 of vagina)** the visceral afferents for pain travel via the **parasympathetics**
 - They travel in the inferior hypogastric plexus
 - Pain is referred to the **S2-S4** areas



Pelvic pain line

- The concept of the "pelvic pain line" is often used in anatomy and medicine to understand the patterns of pain sensation in the pelvis. It's essentially a boundary within the pelvis that separates areas with different types of sensory innervation:
 - **Above the Pelvic Pain Line (peritoneal portion of bladder, uterine body, uterine fundus and uterine tubes):** GVA fibers from structures above the pain line follow the course of the sympathetic efferent fibers. **Pain is referred to the T10-L2 areas**
 - **Below the Pelvic Pain Line (majority of the urinary bladder, cervix and upper 2/3 of vagina):** GVA fibers from structures below the pain line follow the course of the parasympathetic efferent. **Pain is referred to the S2-S4 areas**